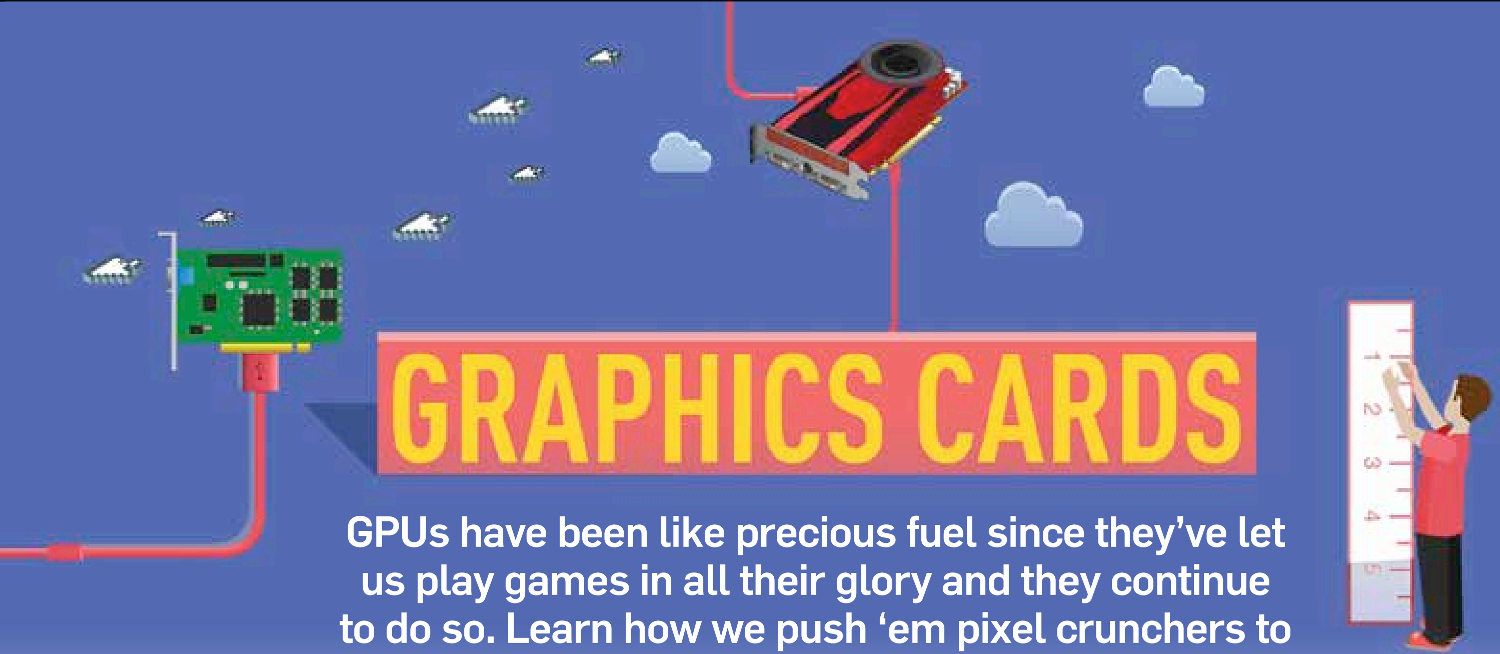

GRAPHICS CARDS

GPUs have been like precious fuel since they've let us play games in all their glory and they continue to do so. Learn how we push 'em pixel crunchers to their absolute limit when we test them.

By **Mithun Mohandas**

Very few PC components are as exciting as graphics cards to tinker with, after all you do get to play an awful lot of games under the pretext of "testing". However, graphics cards have come a long way since the early days when hardware roles were more discrete. These days we have CPUs and GPUs which have adopted functions which are just shy of being truly homogenous. We'll debate on this later, for now, let's get you started with benchmarking these pixel-crunching wonders.

Types

For starters, graphics cards are bifurcated into two groups because the computational workload performed for these tasks are vastly different. Moreover, the number of users who perform both of these tasks are quite minimal. This is why, having a one-chip solution that caters to both usage scenarios wouldn't bode well since most of the users will end up paying for something that they're never going to use. So which are these two distinct groups? Well, the overbearing majority happens to be us gamers and the minority (with deep pockets) are the ones that dabble in graphics content creation.

There is a third class which we haven't spoken about and will not be covering for obvious reasons – the supercomputer chips (for those with endlessly deep pockets).

Gaming

These are the cards that most of us have heard of. There are two competitors as far as discrete cards are concerned – AMD and NVIDIA. Intel has a pretty decent graphics solution which is integrated into a good number of CPUs in their lineup but they come nowhere to the capabilities of these discrete cards. Gaming cards are tasked with pushing as many FPS (Frames Per Second) in games as they can. The more expensive the card is, the more FPS you are going to get. And as the years pass, these high-priced cards retain the capability to render whatever the current game engines can throw at them while the more economically priced cards stutter at the mere mention of a AAA title. Implementing proprietary APIs is another thing that these cards do and each competitor has a good bunch of 'em which work wonderfully on their offerings while rendering the cards from the competition lethargic. NVIDIA's lineup goes by the GeForce moniker while AMD uses the Radeon label.